Lectures on Partial Extrinsic Curvature (Draft 2)

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Introduction

The Theory of Partial Extrinsic Curvature (PXC) is a geometric toy theory of gravity based on the concept of extrinsic curvature. In PXC, we have two coordinate systems; the flat container space and the warped embedded space. The container space is unobservable, but the "embedding" makes up all of space as we experience it, like an aether built of the background itself.

Lecture 1 will be discussing the basics of the embedding and it's kinematics. We'll also do a proof-of-concept run to reproduce the Schwarzschild line element and will use it to get Newtonian gravity from the embedding's fictitious forces.

Lecture 2 will derive PXC's field equations from unitary diffeomorphism invariance, constructing the Lavelle tensor and scalar to create a Lagrangian action.

While we will be using Einstein summation convention, we will explicitly **not** have a metric to raise and lower indices; for lecture 1, index placement will be purely notational.

Lecture 1: Embedding Kinematics & Dynamics

The Fundamental Relations

To begin, let's define the zeroth-order "fundamental relation:"

$$dx^i = \phi^i_j dC^j \qquad \phi^i_j = \frac{\partial x^i}{\partial C_j}$$

For now, we can imagine x^i and C_j to be real; this won't, however, be the case in practice. This relation then tells us that dimensions are mixed when transformed into embedding coordinates; every component dx^i does not depend only on it's counterpart dC^i but instead on contributions from every direction weighted by ϕ^i_j.

Defining that $x^4 = t$ is the experienced time in the embedding, we can divide through to get the first-order fundamental relation:

$$\dot{x}^i = \phi^i_j \dot{C}^j$$

Differentiating once gives the second-order fundamental relation:

$$\ddot{x}^i = \frac{\partial\left(\phi^i_j \dot{C}^j\right)}{\partial t} = \frac{\partial \phi^i_j}{\partial t}\dot{C}^j + \phi^i_j \ddot{C}^j$$

We can chain rule the $\partial\phi^i_j/\partial t$ derivative:

$$\frac{\partial \phi^i_j}{\partial t} = \frac{\partial \phi^i_j}{\partial C_k}\dot{C}^k \implies \ddot{x}^i = \frac{\partial \phi^i_j}{\partial C_k}\dot{C}^j\dot{C}^k + \phi^i_j\ddot{C}^j = \psi^i_{jk}\dot{C}^j\dot{C}^k + \phi^i_j\ddot{C}^j \qquad \psi^i_{jk} = \frac{\partial x^i}{\partial C_j \partial C_k}$$

This is the fundamental relation at second order. We have two terms to discuss here:

1. $\psi^i_{jk}\dot{C}^j\dot{C}^k$ is a fictitious force sourced by the curvature of the embedding. Intuitively, we can imagine a straight line on a curved graph appearing to diverge from straight in graph coordinates. An embedding with no second order change will not generate this term; curvature, second order change, is required to create it.
2. $\phi^i_j\ddot{C}^j$ is the transformation of incoming acceleration into embedded coordinates, using the same method the first order fundamental relation uses to convert incoming velocities into embedding coordinates. Again, like the zeroth order fundamental relation, accelerations are mixed as their transformed.

Thus, the final set of fundamental relations read:

$$dx^i = \phi^i_j dC^j \qquad \dot{x}^i = \phi^i_j \dot{C}^j \qquad \ddot{x}^i = \psi^i_{jk}\dot{C}^j\dot{C}^k + \phi^i_j\ddot{C}^j$$

The PXC Field Equations

To finish up our purely theoretical section, we'll derive our field equations from diffeomorphism invariance. In PXC, because the container is unobservable, we don't want coordinate changes in it to affect the embedding's curvature; thus we want invariance under the diffeomorphism $C \to C'$.

When the integration measure transforms under the diffeomorphism $C \to C'$, we get:

$$d^4C \to d^4C \left|\det \frac{\partial C'^i}{\partial C_j}\right| = \frac{d^4C}{|\det \gamma|} \qquad \gamma^i_j = \frac{\partial C^i}{\partial C'_j}$$

When ϕ transforms, we get:

$$\phi^i_j = \frac{\partial x^i}{\partial C_j} \to \frac{\partial x^i}{\partial C'_j} = \frac{\partial x^i}{\partial C_k}\frac{\partial C^k}{\partial C'_j} = \phi^i_k \gamma^k_j$$

So when ψ transforms, we get:

$$\psi^i_{jk} = \frac{\partial \phi^i_j}{\partial C_k} \to \frac{\partial \phi'^i_j}{\partial C'_k} = \frac{\partial(\phi^i_m \gamma^m_j)}{\partial C_n}\gamma^n_k = \left(\frac{\partial \phi^i_m}{\partial C_n}\gamma^m_j + \phi^i_m \frac{\partial \gamma^m_j}{\partial C_n}\right)\gamma^n_k = \psi^i_{mn}\gamma^m_j\gamma^n_k + \phi^i_m \omega^m_{jn}\gamma^n_k$$

where $\omega^m_{jn} = \partial\gamma^m_j/\partial C_n$. ψ Is not invariant under diffeomorphisms; thus we need to build some object in which the extra terms cancel. Defining $\chi = \phi^{-1}$, let's create an object called the "Lavelle symbol":

$$\mathscr{L}^i_{jk} = \chi^i_m \psi^m_{jk}$$

Note that this is symmetric in it's lower indices. We can see how it transforms under the diffeomorphism $C \to C'$ with $\eta = \gamma^{-1}$:

$$\mathscr{L}^i_{jk} \to \eta^i_n \chi^n_m \left(\psi^m_{ab}\gamma^a_j\gamma^b_k + \phi^m_a \omega^a_{jb}\gamma^b_k\right) = \eta^i_n\chi^n_m\psi^m_{ab}\gamma^a_j\gamma^b_k + \eta^i_n\chi^n_m\phi^m_a\omega^a_{jb}\gamma^b_k$$

$\chi\phi = I$, so:

$$\mathscr{L}^i_{jk} \to \eta^i_n\mathscr{L}^n_{ab}\gamma^a_j\gamma^b_k + \eta^i_n\delta^n_a\omega^a_{jb}\gamma^b_k = \eta^i_d\mathscr{L}^d_{ab}\gamma^a_j\gamma^b_k + \eta^i_d\omega^d_{jb}\gamma^b_k$$

which is the standard connection transformation law:

$$\Gamma^i_{jk} \to \frac{\partial x'^i}{\partial x^m}\frac{\partial x^a}{\partial x'^j}\frac{\partial x^b}{\partial x'^k}\Gamma^m_{ab} + \frac{\partial x'^i}{\partial x^m}\frac{\partial^2 x^m}{\partial x'^j \partial x'^k}$$

with $\eta^i_d = \partial x'^i/\partial x^d$, $\gamma^a_j = \partial x^a/\partial x'^j$, and $\omega^d_{jb}\gamma^b_k = \partial^2 x^d/\partial x'^j\partial x'^k$. Thus, we'll use a Riemann tensor:

$$R^r_{smn} = \frac{\partial\mathscr{L}^r_{ns}}{\partial C_m} - \frac{\partial\mathscr{L}^r_{ms}}{\partial C_n} + \mathscr{L}^r_{ml}\mathscr{L}^l_{ns} - \mathscr{L}^r_{nl}\mathscr{L}^l_{ms}$$

Under diffeomorphism, we get:

$$R^r_{smn} \to \eta^r_a\gamma^b_s\gamma^c_m\gamma^d_n R^a_{bcd}$$

Taking the trace over r and m pairs η and γ:

$$R^r_{srn} \to \gamma^b_s\gamma^d_n R^a_{bcd}\cdot\eta^r_a\gamma^c_r = \gamma^b_s\gamma^d_n R^a_{bcd}\delta^c_a = \gamma^b_s\gamma^d_n R^c_{bcd}$$

We need a scalar that peels off two γs; we can try $\chi\chi R$:

$$\chi^s_u\chi^n_v R^r_{srn} \to \eta^s_a\chi^a_u\eta^n_m\chi^m_v\gamma^b_s\gamma^d_n R^c_{bcd} = \chi^a_u\chi^m_v\delta^b_a\delta^d_m R^c_{bcd} = \chi^a_u\chi^m_v R^c_{acm} = \Omega_{uv}$$

Ω is invariant under diffeomorphisms. Taking the trace over u, v and using $\det\phi$ to counteract the $1/|\det\gamma|$ in the measure, we get the PXC action:

$$S = \int d^4C\, L = \int d^4C\, |\det\phi|\,\Omega$$

$$\Omega = \Omega_{aa} = \Upsilon^{sn}R^r_{srn} \qquad \Upsilon^{sn} = \chi^s_a\chi^n_a \qquad R^r_{smn} = \frac{\partial\mathscr{L}^r_{ns}}{\partial C_m} - \frac{\partial\mathscr{L}^r_{ms}}{\partial C_n} + \mathscr{L}^r_{ml}\mathscr{L}^l_{ns} - \mathscr{L}^r_{nl}\mathscr{L}^l_{ms}$$

$$\mathscr{L}^i_{jk} = \chi^i_m\psi^m_{jk}$$

Euler-Lagrange I

To vary this, we'll use the Euler-Lagrange equations:

$$0 = \frac{\partial L}{\partial \phi^i_j} - \frac{\partial}{\partial C^k}\frac{\partial L}{\partial \psi^i_{jk}}$$

First, the ϕ^i_j term:

$$\frac{\partial L}{\partial \phi^i_j} = \frac{\partial(|\det\phi|\Omega)}{\partial \phi^i_j} = \frac{\partial|\det\phi|}{\partial \phi^i_j}\Omega + |\det\phi|\frac{\partial\Omega}{\partial\phi^i_j}$$

Taking each piece:

$$\frac{\partial|\det\phi|}{\partial\phi^i_j} = |\det\phi|\,\chi^j_i \qquad \frac{\partial\Omega}{\partial\phi^i_j} = \frac{\partial(\Upsilon^{sn}R^r_{srn})}{\partial\phi^i_j} = \frac{\partial\Upsilon^{sn}}{\partial\phi^i_j}R^r_{srn} + \Upsilon^{sn}\frac{\partial R^r_{srn}}{\partial\phi^i_j}$$

Taking the Υ^{sn} term first:

$$\frac{\partial\Upsilon^{sn}}{\partial\phi^i_j} = \frac{\partial(\chi^s_a\chi^n_a)}{\partial\phi^i_j} = \frac{\partial\chi^s_n}{\partial\phi^i_j}\chi^n_a + \chi^s_n\frac{\partial\chi^n_a}{\partial\phi^i_j} = -\chi^s_i\chi^j_a\chi^n_a + \chi^s_a\chi^j_a\chi^j_a = -\chi^s_i\Upsilon^{jn} - \chi^n_i\Upsilon^{sj}$$

Then the R^r_{srn} term:

$$R^r_{srn} = \frac{\partial\mathscr{L}^r_{ns}}{\partial C_r} - \frac{\partial\mathscr{L}^r_{rs}}{\partial C_n} + \mathscr{L}^r_{rl}\mathscr{L}^l_{ns} - \mathscr{L}^r_{nl}\mathscr{L}^l_{rs}$$

$$\implies \frac{\partial R^r_{srn}}{\partial\phi^i_j} = \frac{\partial^2\mathscr{L}^r_{ns}}{\partial C_r\partial\phi^i_j} - \frac{\partial^2\mathscr{L}^r_{rs}}{\partial C_n\partial\phi^i_j} + \frac{\partial(\mathscr{L}^r_{rl}\mathscr{L}^l_{ns})}{\partial\phi^i_j} - \frac{\partial(\mathscr{L}^r_{nl}\mathscr{L}^l_{rs})}{\partial\phi^i_j}$$

Again taking the pieces:

$$\frac{\partial\mathscr{L}^r_{ns}}{\partial\phi^i_j} = \frac{\partial(\chi^r_m\psi^m_{ns})}{\partial\phi^i_j} = \psi^m_{ns}\frac{\partial\chi^r_m}{\partial\phi^i_j} = -\psi^m_{ns}\chi^r_i\chi^j_m = -\chi^r_i\mathscr{L}^j_{ns}$$

$$\frac{\partial^2\mathscr{L}^r_{ns}}{\partial C_r\partial\phi^i_j} = \frac{\partial(-\chi^r_i\mathscr{L}^j_{ns})}{\partial C_r} = -\frac{\partial\chi^r_i}{\partial C_r}\mathscr{L}^j_{ns} - \chi^r_i\frac{\partial\mathscr{L}^j_{ns}}{\partial C_r} \qquad \frac{\partial\chi^r_i}{\partial C_r} = \frac{\partial\chi^r_i}{\partial\phi^a_b}\psi^a_{br} = -\chi^r_a\chi^b_i\psi^a_{br} = -\chi^b_i\mathscr{L}^r_{br}$$

$$\frac{\partial^2\mathscr{L}^r_{ns}}{\partial C_r\partial\phi^i_j} = \chi^b_i\mathscr{L}^r_{br}\mathscr{L}^j_{ns} - \chi^r_i\frac{\partial\mathscr{L}^j_{ns}}{\partial C_r} \qquad \frac{\partial^2\mathscr{L}^r_{rs}}{\partial C_n\partial\phi^i_j} = \chi^b_i\mathscr{L}^r_{bn}\mathscr{L}^j_{rs} - \chi^r_i\frac{\partial\mathscr{L}^j_{rs}}{\partial C_n}$$

Combined:

$$\frac{\partial^2\mathscr{L}^r_{ns}}{\partial C_r\partial\phi^i_j} - \frac{\partial^2\mathscr{L}^r_{rs}}{\partial C_n\partial\phi^i_j} = \chi^b_i\mathscr{L}^r_{br}\mathscr{L}^j_{ns} - \chi^r_i\frac{\partial\mathscr{L}^j_{ns}}{\partial C_r} - \chi^b_i\mathscr{L}^r_{bn}\mathscr{L}^j_{rs} + \chi^r_i\frac{\partial\mathscr{L}^j_{rs}}{\partial C_n}$$

$$\frac{\partial^2\mathscr{L}^r_{ns}}{\partial C_r\partial\phi^i_j} - \frac{\partial^2\mathscr{L}^r_{rs}}{\partial C_n\partial\phi^i_j} = \chi^r_i\left(\frac{\partial\mathscr{L}^j_{rs}}{\partial C_n} - \frac{\partial\mathscr{L}^j_{ns}}{\partial C_r}\right) + \chi^b_i\left(\mathscr{L}^r_{br}\mathscr{L}^j_{ns} - \mathscr{L}^r_{bn}\mathscr{L}^j_{rs}\right)$$

$$\frac{\partial^2 \mathscr{L}^r_{ns}}{\partial C_r \partial \phi^i_j} - \frac{\partial^2 \mathscr{L}^r_{rs}}{\partial C_n \partial \phi^i_j} = \chi^b_i \left(\frac{\partial \mathscr{L}^j_{bs}}{\partial C_n} - \frac{\partial \mathscr{L}^j_{ns}}{\partial C_b} + \mathscr{L}^r_{br} \mathscr{L}^j_{ns} - \mathscr{L}^r_{bn} \mathscr{L}^j_{rs} \right)$$

Notice that this is just negative R:

$$\frac{\partial^2 \mathscr{L}^r_{ns}}{\partial C_r \partial \phi^i_j} - \frac{\partial^2 \mathscr{L}^r_{rs}}{\partial C_n \partial \phi^i_j} = -\chi^b_i R^j_{sbn}$$

Next, the quadratic terms:

$$\frac{\partial(\mathscr{L}^r_{rl}\mathscr{L}^l_{ns})}{\partial \phi^i_j} = \frac{\partial(\chi^r_m \psi^m_{rl} \chi^l_u \psi^u_{ns})}{\partial \phi^i_j} = \psi^m_{rl}\psi^u_{ns} \frac{\partial(\chi^r_m \chi^l_u)}{\partial \phi^i_j} = \psi^m_{rl}\psi^u_{ns} \left(\frac{\partial \chi^r_m}{\partial \phi^i_j} \chi^l_u + \chi^r_m \frac{\partial \chi^l_u}{\partial \phi^i_j} \right)$$

$$\frac{\partial(\mathscr{L}^r_{rl}\mathscr{L}^l_{ns})}{\partial \phi^i_j} = -\psi^m_{rl}\psi^u_{ns} \left(\chi^r_i \chi^j_m \chi^l_u + \chi^r_m \chi^l_i \chi^j_u \right) = -\psi^m_{rl}\psi^u_{ns}\chi^r_i\chi^j_m\chi^l_u - \psi^m_{rl}\psi^u_{ns}\chi^r_m\chi^l_i\chi^j_u$$

$$\frac{\partial(\mathscr{L}^r_{rl}\mathscr{L}^l_{ns})}{\partial \phi^i_j} = -\chi^r_i \mathscr{L}^j_{rl}\mathscr{L}^l_{ns} - \chi^l_i \mathscr{L}^r_{rl}\mathscr{L}^j_{ns} \qquad \frac{\partial(\mathscr{L}^r_{nl}\mathscr{L}^l_{rs})}{\partial \phi^i_j} = -\chi^r_i \mathscr{L}^j_{nl}\mathscr{L}^l_{rs} - \chi^l_i \mathscr{L}^r_{nl}\mathscr{L}^j_{rs}$$

Taking the difference:

$$\frac{\partial(\mathscr{L}^r_{rl}\mathscr{L}^l_{ns})}{\partial \phi^i_j} - \frac{\partial(\mathscr{L}^r_{nl}\mathscr{L}^l_{rs})}{\partial \phi^i_j} = -\chi^r_i \mathscr{L}^j_{rl}\mathscr{L}^l_{ns} - \chi^l_i \mathscr{L}^r_{rl}\mathscr{L}^j_{ns} + \chi^r_i \mathscr{L}^j_{nl}\mathscr{L}^l_{rs} + \chi^l_i \mathscr{L}^r_{nl}\mathscr{L}^j_{rs}$$

$$\frac{\partial(\mathscr{L}^r_{rl}\mathscr{L}^l_{ns})}{\partial \phi^i_j} - \frac{\partial(\mathscr{L}^r_{nl}\mathscr{L}^l_{rs})}{\partial \phi^i_j} = \chi^r_i \left(\mathscr{L}^j_{nl}\mathscr{L}^l_{rs} - \mathscr{L}^j_{rl}\mathscr{L}^l_{ns} + \mathscr{L}^l_{nr}\mathscr{L}^j_{ls} - \mathscr{L}^l_{rl}\mathscr{L}^j_{rs} \right)$$

These are the quadratic pieces to R^j_{snr} and R^j_{srn} respectively. Together, $\partial R^r_{srn}/\partial \phi^i_j$ is:

$$\frac{\partial R^r_{srn}}{\partial \phi^i_j} = \chi^r_i \left(-R^j_{srn} + \mathscr{L}^j_{nl}\mathscr{L}^l_{rs} - \mathscr{L}^j_{rl}\mathscr{L}^l_{ns} + \mathscr{L}^l_{nr}\mathscr{L}^j_{ls} - \mathscr{L}^l_{rl}\mathscr{L}^j_{rs} \right)$$

after relabeling $b \to r$. Then $\partial\Omega/\partial\phi^i_j$ is:

$$\frac{\partial \Omega}{\partial \phi^i_j} = -(\chi^s_i \Upsilon^{jn} + \chi^n_i \Upsilon^{sj}) R^r_{srn} + \Upsilon^{sn} \chi^r_i \left(-R^j_{srn} + \mathscr{L}^j_{nl}\mathscr{L}^l_{rs} - \mathscr{L}^j_{rl}\mathscr{L}^l_{ns} + \mathscr{L}^l_{nr}\mathscr{L}^j_{ls} - \mathscr{L}^l_{rl}\mathscr{L}^j_{rs} \right)$$

and $\partial L/\partial\phi^i_j$ is:

$$\frac{\partial L}{\partial \phi^i_j} = |\det \phi| \begin{pmatrix} +\chi^j_i \Omega \\ -(\chi^s_i \Upsilon^{jn} + \chi^n_i \Upsilon^{sj}) R^r_{srn} \\ +\Upsilon^{sn} \chi^r_i \left(-R^j_{srn} + \mathscr{L}^j_{nl}\mathscr{L}^l_{rs} - \mathscr{L}^j_{rl}\mathscr{L}^l_{ns} + \mathscr{L}^l_{nr}\mathscr{L}^j_{ls} - \mathscr{L}^l_{rl}\mathscr{L}^j_{rs} \right) \end{pmatrix}$$

Since Υ^{jn} is symmetric in j, n, $\chi^s_i \Upsilon^{jn} + \chi^n_i \Upsilon^{sj} = 2\chi^s_i \Upsilon^{jn}$:

$$\frac{\partial L}{\partial \phi^i_j} = |\det \phi| \begin{pmatrix} +\chi^j_i \Omega - 2\chi^s_i \Upsilon^{jn} R^r_{srn} \\ +\Upsilon^{sn} \chi^r_i \left(-R^j_{srn} + \mathscr{L}^j_{nl}\mathscr{L}^l_{rs} - \mathscr{L}^j_{rl}\mathscr{L}^l_{ns} + \mathscr{L}^l_{nr}\mathscr{L}^j_{ls} - \mathscr{L}^l_{rl}\mathscr{L}^j_{rs} \right) \end{pmatrix}$$

Expanding out the bottom piece:

$$\begin{aligned} -R^j_{srn} + \mathscr{L}^j_{nl}\mathscr{L}^l_{rs} - \mathscr{L}^j_{rl}\mathscr{L}^l_{ns} + \mathscr{L}^l_{nr}\mathscr{L}^j_{ls} - \mathscr{L}^l_{rl}\mathscr{L}^j_{rs} = & -\frac{\partial \mathscr{L}^j_{ns}}{\partial C_r} + \frac{\partial \mathscr{L}^j_{rs}}{\partial C_n} \\ & - \mathscr{L}^j_{rl}\mathscr{L}^l_{ns} + \mathscr{L}^j_{nl}\mathscr{L}^l_{rs} \\ & + \mathscr{L}^j_{nl}\mathscr{L}^l_{rs} - \mathscr{L}^j_{rl}\mathscr{L}^l_{ns} \\ & + \mathscr{L}^l_{nr}\mathscr{L}^j_{ls} - \mathscr{L}^l_{rl}\mathscr{L}^j_{rs} \end{aligned}$$

$$\begin{aligned} & -\frac{\partial \mathscr{L}^j_{ns}}{\partial C_r} + \frac{\partial \mathscr{L}^j_{rs}}{\partial C_n} \\ & -\mathscr{L}^j_{rl}\mathscr{L}^l_{ns} + \mathscr{L}^j_{nl}\mathscr{L}^l_{rs} = \frac{\partial \mathscr{L}^j_{rs}}{\partial C_n} - \frac{\partial \mathscr{L}^j_{ns}}{\partial C_r} + \mathscr{L}^l_{nr}\mathscr{L}^j_{ls} - \mathscr{L}^l_{rl}\mathscr{L}^j_{rs} + 2\mathscr{L}^j_{nl}\mathscr{L}^l_{rs} - 2\mathscr{L}^j_{rl}\mathscr{L}^l_{ns} \\ & +\mathscr{L}^j_{nl}\mathscr{L}^l_{rs} - \mathscr{L}^j_{rl}\mathscr{L}^l_{ns} \\ & +\mathscr{L}^l_{nr}\mathscr{L}^j_{ls} - \mathscr{L}^l_{rl}\mathscr{L}^j_{rs} \end{aligned}$$

which is $R^j_{snr} - 2\left(\mathscr{L}^j_{nl}\mathscr{L}^l_{rs} - \mathscr{L}^j_{rl}\mathscr{L}^l_{ns}\right)$, so the expression for $\partial L/\partial \phi^i_j$ is:

$$\frac{\partial L}{\partial \phi^i_j} = \chi^j_i \Omega - 2\chi^s_i \Upsilon^{jn} R^r_{srn} + \Upsilon^{sn}\chi^r_i R^j_{snr} - 2\Upsilon^{sn}\chi^r_i \left(\mathscr{L}^j_{nl}\mathscr{L}^l_{rs} - \mathscr{L}^j_{rl}\mathscr{L}^l_{ns}\right)$$

Notice that the bracket is antisymmetric in n, s where Υ is symmetric in n, s, so that contraction is 0 and we get a final expression of:

$$\frac{\partial L}{\partial \phi^i_j} = \chi^j_i \Omega - 2\chi^s_i \Upsilon^{jn} R^r_{srn} + \Upsilon^{sn}\chi^r_i R^j_{snr}$$

Euler-Lagrange II

The next derivative is:

$$-\frac{\partial}{\partial C^k}\frac{\partial L}{\partial \psi^i_{jk}}$$

We can rewrite this in terms of the Lavelle symbol:

$$-\frac{\partial}{\partial C^k}\frac{\partial L}{\partial \psi^i_{jk}} = -\frac{\partial}{\partial C_k}\frac{\partial L}{\partial \mathscr{L}^a_{bc}}\frac{\partial \mathscr{L}^a_{bc}}{\partial \psi^i_{jk}} \qquad \frac{\partial \mathscr{L}^a_{bc}}{\partial \psi^i_{jk}} = \frac{\partial(\chi^a_m \psi^m_{bc})}{\partial \psi^i_{jk}} = \chi^a_m \frac{\partial \psi^m_{bc}}{\partial \psi^i_{jk}} = \chi^a_m \delta^{im}\delta_{jb}\delta_{ck}$$

So:

$$-\frac{\partial}{\partial C^k}\frac{\partial L}{\partial \psi^i_{jk}} = -\frac{\partial}{\partial C_k}\left(\frac{\partial L}{\partial \mathscr{L}^a_{bc}}\chi^a_m \delta^{im}\delta_{jb}\delta_{ck}\right) = -\frac{\partial}{\partial C_k}\left(\frac{\partial L}{\partial \mathscr{L}^a_{jk}}\chi^a_i\right)$$

Since $L = |\det \phi|\Omega = |\det \phi|\Upsilon^{sn}R^r_{srn}$ and neither $|\det \phi|$ or Υ^{sn} depend on ψ:

$$\frac{\partial L}{\partial \mathscr{L}^a_{jk}} = |\det \phi|\Upsilon^{sn}\frac{\partial R^r_{srn}}{\partial \mathscr{L}^a_{jk}} \qquad \frac{\partial R^r_{srn}}{\partial \mathscr{L}^a_{jk}} = \frac{\partial^2 \mathscr{L}^r_{ns}}{\partial C_r \partial \mathscr{L}^a_{jk}} - \frac{\partial^2 \mathscr{L}^r_{rs}}{\partial C_n \partial \mathscr{L}^a_{jk}} + \frac{\partial\left(\mathscr{L}^r_{rl}\mathscr{L}^l_{ns}\right)}{\partial \mathscr{L}^a_{jk}} - \frac{\partial(\mathscr{L}^r_{nl}\mathscr{L}^l_{rs})}{\partial \mathscr{L}^a_{jk}}$$

$\partial\mathscr{L}/\partial\mathscr{L} \sim \delta$ and the C derivatives of δs are zero, so the first two terms vanish; we can use the product rule on the last two:

$$\frac{\partial\left(\mathscr{L}^r_{rl}\mathscr{L}^l_{ns}\right)}{\partial\mathscr{L}^a_{jk}} - \frac{\partial(\mathscr{L}^r_{nl}\mathscr{L}^l_{rs})}{\partial\mathscr{L}^a_{jk}} = \frac{\partial\mathscr{L}^r_{rl}}{\partial\mathscr{L}^a_{jk}}\mathscr{L}^l_{ns} + \mathscr{L}^r_{rl}\frac{\partial\mathscr{L}^l_{ns}}{\partial\mathscr{L}^a_{jk}} - \frac{\partial\mathscr{L}^r_{nl}}{\partial\mathscr{L}^a_{jk}}\mathscr{L}^l_{rs} - \mathscr{L}^r_{nl}\frac{\partial\mathscr{L}^l_{rs}}{\partial\mathscr{L}^a_{jk}}$$

Taking the derivatives:

$$\frac{\partial\left(\mathscr{L}^r_{rl}\mathscr{L}^l_{ns}\right)}{\partial\mathscr{L}^a_{jk}} - \frac{\partial(\mathscr{L}^r_{nl}\mathscr{L}^l_{rs})}{\partial\mathscr{L}^a_{jk}} = \delta^{ra}\delta_{rj}\delta_{lk}\mathscr{L}^l_{ns} + \mathscr{L}^r_{rl}\delta^{la}\delta_{nj}\delta_{sk} - \delta^{ra}\delta_{jn}\delta_{kl}\mathscr{L}^l_{rs} - \mathscr{L}^r_{nl}\delta^{al}\delta_{rj}\delta_{ks}$$

Collapsing the δs:

$$\frac{\partial\left(\mathscr{L}^r_{rl}\mathscr{L}^l_{ns}\right)}{\partial\mathscr{L}^a_{jk}} - \frac{\partial(\mathscr{L}^r_{nl}\mathscr{L}^l_{rs})}{\partial\mathscr{L}^a_{jk}} = \delta^a_j\mathscr{L}^k_{ns} + \mathscr{L}^r_{ra}\delta_{nj}\delta_{sk} - \delta_{jn}\mathscr{L}^k_{as} - \mathscr{L}^j_{na}\delta_{ks}$$

So:

$$\frac{\partial R^r_{srn}}{\partial\mathscr{L}^a_{jk}} = \delta^a_j\mathscr{L}^k_{ns} + \mathscr{L}^r_{ra}\delta_{nj}\delta_{sk} - \delta_{jn}\mathscr{L}^k_{as} - \mathscr{L}^j_{na}\delta_{ks}$$

$$\frac{\partial L}{\partial\mathscr{L}^a_{jk}} = |\det\phi|\Upsilon^{sn}\left(\delta^a_j\mathscr{L}^k_{ns} + \mathscr{L}^r_{ra}\delta_{nj}\delta_{sk} - \delta_{jn}\mathscr{L}^k_{as} - \mathscr{L}^j_{na}\delta_{ks}\right)$$

Contracting with Υ^{sn}, then χ^a_i:

$$\frac{\partial L}{\partial\mathscr{L}^a_{jk}} = |\det\phi|\left(\delta^a_j\mathscr{L}^k_{ns}\Upsilon^{sn} + \mathscr{L}^r_{ra}\Upsilon^{jk} - \mathscr{L}^k_{as}\Upsilon^{sj} - \mathscr{L}^j_{na}\Upsilon^{kn}\right)$$

$$\frac{\partial L}{\partial\mathscr{L}^a_{jk}}\chi^a_i = |\det\phi|\left(\chi^j_i\mathscr{L}^k_{ns}\Upsilon^{sn} + \chi^a_i\mathscr{L}^r_{ra}\Upsilon^{jk} - \chi^a_i\mathscr{L}^k_{as}\Upsilon^{sj} - \chi^a_i\mathscr{L}^j_{na}\Upsilon^{kn}\right)$$

Taking $\partial/\partial C^k$:

$$\frac{\partial}{\partial C_k}\left(\frac{\partial L}{\partial\mathscr{L}^a_{jk}}\chi^a_i\right) = \frac{\partial}{\partial C_k}\left(|\det\phi|\left(\chi^j_i\mathscr{L}^k_{ns}\Upsilon^{sn} + \chi^a_i\mathscr{L}^r_{ra}\Upsilon^{jk} - \chi^a_i\mathscr{L}^k_{as}\Upsilon^{sj} - \chi^a_i\mathscr{L}^j_{na}\Upsilon^{kn}\right)\right)$$

This would be unwieldy to expand, so we'll leave it compact for now. The final field equation is:

$$0 = \begin{array}{l}\chi^j_i\Omega - 2\chi^s_i\Upsilon^{jn}R^r_{srn} + \Upsilon^{sn}\chi^r_iR^j_{snr} \\ \dfrac{\partial}{\partial C_k}\left(|\det\phi|\left(\chi^j_i\mathscr{L}^k_{ns}\Upsilon^{sn} + \chi^a_i\mathscr{L}^r_{ra}\Upsilon^{jk} - \chi^a_i\mathscr{L}^k_{as}\Upsilon^{sj} - \chi^a_i\mathscr{L}^j_{na}\Upsilon^{kn}\right)\right)\end{array}$$